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RESEARCH ARTICLE

Parental herbivory shapes offspring performance across genotypes in an invasive clonal plant

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Keywords
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ABSTRACT

- Parental environmental effects play a crucial role in clonal plants, potentially enabling clonal offspring to pre-adapt to environmental changes. However, few studies have examined whether such parental effects differ between genotypes and whether they enhance both growth and defence.
- We conducted a two-generation herbivory experiment using six genotypes of the invasive clonal plant *Hydrocotyle verticillata* and the native generalist herbivore *Spodoptera litura* in China. We tested how parental herbivory (PH) effects influence offspring survival, growth, and herbivory defence, and whether these parental effects vary among genotypes.
- Herbivory significantly reduced leaf mass and leaf number of parental plants, while increasing its specific leaf area. PH enhanced offspring survival by 21% and tended to increase offspring leaf mass. However, clonal offspring from herbivory-treated parental plants suffered equally severe defoliation (>95%) as control offspring. While most parental effects were consistent across genotypes, root mass and specific leaf area showed genotype-dependent responses. Notably, genotype G4 exhibited markedly elevated SLA following PH despite having the lowest offspring survival.
- Our results reveal an asymmetric transgenerational strategy in invasive *H. verticillata*, that is, PH enhances offspring growth and survival without conferring herbivory defence. This decoupling of growth enhancement from defence induction represents an evolutionary trade-off that may facilitate rapid colonization but leaves invasive populations vulnerable to herbivore pressure. Consequently, while this transgenerational plasticity facilitates range expansion and population establishment in herbivore-free habitats, it may render invasive populations susceptible to regulation by specialist or generalist herbivores during secondary contact.

INTRODUCTION

Plant invasions are a primary driver of global environmental change, causing native biodiversity loss and ecosystem degradation (Pyšek *et al.* 2020; IPBES 2023). The invasion success of alien plants is often explained by the Enemy Release Hypothesis (ERH), which posits that invasive plants gain competitive advantages by escaping their co-evolved specialist herbivores, allowing resource reallocation from defence to growth and reproduction (Keane 2002; Liu & Stiling 2006; Xu *et al.* 2021). This evolutionary realignment may leave invasive populations particularly vulnerable to novel herbivore encounters, but paradoxically, many invasive species thrive despite herbivory from generalist herbivores in their introduced ranges (Huang *et al.* 2010; Schaffner *et al.* 2011). Understanding how invasive plants maintain advantages under herbivore pressure without their ancestral defensive mechanisms remains a critical gap in invasion ecology.

Transgenerational plasticity may provide a rapid adaptive mechanism for invasive plants facing novel herbivore communities (Agrawal 2001; Sobral *et al.* 2021; Fu *et al.* 2023). Parental environmental effects, whereby parental stress exposure modifies offspring phenotypes, can enable populations to respond to environmental challenges faster than genetic adaptation alone (Engqvist & Reinhold 2016; Yin *et al.* 2019). In clonal plants, these effects may be particularly pronounced as vegetative reproduction maintains direct physiological connections between generations (Dong *et al.* 2018a; Luo *et al.* 2022). Such transgenerational priming could theoretically compensate for lost defensive capabilities by enhancing offspring growth vigor or inducing anticipatory defences. However, the effectiveness of transgenerational plasticity in conferring herbivory defence remains largely untested in invasive contexts, particularly when invasive populations have undergone enemy release.

Herbivory represents one of the most prevalent and ecologically significant stresses for plants, directly constraining growth through tissue loss and resource depletion (Schoonhoven *et al.* 2005). Recent evidence suggests herbivory effects can persist across generations, with parental herbivory (PH) either

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compromising offspring through reduced provisioning or enhancing offspring performance through adaptive priming (Herman & Sultan 2011; Holeski *et al.* 2012). In invasive populations that have evolved reduced constitutive defences, the direction and magnitude of these transgenerational effects become critical: enhanced growth provisioning might sustain rapid population expansion, while defensive priming could compensate for lost defence traits (Fenesi *et al.* 2014; Sobral *et al.* 2021). Yet, whether invasive plants can maintain both growth enhancement and defensive capabilities through transgenerational mechanisms remains unknown.

Genetic variation fundamentally shapes both plastic responses and their transgenerational transmission (Lavergne & Molofsky 2007; Forsman 2014). Different genotypes may vary substantially in their capacity for transgenerational plasticity, with some lineages exhibiting strong parental effects while others show minimal responsiveness (Li *et al.* 2016). This genotypic variation in transgenerational responses could maintain population-level resilience even under limited genetic diversity – a common characteristic of invasive populations. However, most studies examining PH effects focus on single genotypes (Dong *et al.* 2017, 2019; Verhoeven *et al.* 2018; Kafle & Wurst 2019), obscuring the potential for genotype-specific strategies that collectively enhance invasion success (Agrawal 2001; Colicchio 2017).

Understanding transgenerational effects in invasive clonal plants is particularly important because these species often establish large populations from limited genetic material, making parental conditioning potentially crucial for invasive population success in novel environments. Therefore, we studied the invasive clonal plant *Hydrocotyle verticillata* Thunb. and its interactions with the generalist herbivore *Spodoptera litura* (Fabricius, 1775) in China to investigate transgenerational responses to herbivory across multiple genotypes. This system also provides an ideal model as *H. verticillata* exhibits substantial genotypic variation despite clonal reproduction and faces herbivory from native generalists in its introduced range (Fu *et al.* 2024). We tested three hypotheses: (1) herbivory reduces growth of parental plants while triggering compensatory responses in leaf traits; (2) PH enhances offspring performance through both improved survival and growth and increased herbivory defence; (3) genotypes differ in their transgenerational responses, with some lineages showing stronger parental effects than others. By examining both growth enhancement and defensive capabilities across genotypes, our study could elucidate whether transgenerational plasticity can compensate for evolutionary trade-offs associated with enemy release.

MATERIALS AND METHODS

Study species

Hydrocotyle verticillata, previously misidentified as *Hydrocotyle vulgaris* L. in China, is a perennial clonal herb native to South and North America and Southern Africa (Miao *et al.* 2011; Wang *et al.* 2020). Introduced to China as an ornamental plant in the 1990s, this species has become invasive across tropical and subtropical regions, spreading predominantly through clonal growth rather than sexual reproduction (Miao *et al.* 2011; Wang *et al.* 2020; Fu *et al.* 2024). In its invaded range, *H. verticillata* experiences herbivory from native generalist insects including *S. litura*, which feed on its leaves and potentially constrain population expansion (Fu *et al.* 2024).

Six genotypes of *H. verticillata* (designated G1–G6) were collected from geographically distinct field populations across China, with an average distance of 184 km between sampling sites (Table S1 and Fig. S1). Genotypic identity was confirmed using amplified fragment length polymorphism (AFLP) following Wang *et al.* (2020). Since 2016, clonal ramets of each genotype have been maintained separately under standardized greenhouse conditions at Taizhou University.

S. litura (Lepidoptera: Noctuidae) is a polyphagous generalist herbivore native to China and now distributed across tropical and subtropical regions globally (Maes 2022; EPPO 2025). Its larvae undergo six instars and feed on diverse host plants including crops and wild species (Zhao *et al.* 2019). In China, *S. litura* has been observed feeding on invasive *H. verticillata*, causing substantial defoliation (Fu *et al.* 2024). Experimental larvae were obtained from Henan Jiyuan Baiyun Industrial Co., Ltd., a commercial supplier specializing in biological control agents (http://www.keyunnpv.cn/). Prior to experiments, larvae were reared on a standardized artificial diet (primarily wheat germ-based) in individual Petri dishes (10 cm diameter) under controlled conditions to ensure uniform size and developmental stage.

The experiment was conducted in a greenhouse at Taizhou University, Zhejiang Province, China, with mean temperature of 22.69 °C ± 0.08 °C and mean photosynthetically active radiation of 357.6 μmol m^{-2} s^{-1}.

Experimental design

Parental generation

The study employed a two-factor factorial design incorporating six genotypes of *H. verticillata* and two herbivory treatments (control *versus* herbivory by *S. litura*; Fig. 1A), with 24 replicates per genotype × treatment combination.

On July 4, 2022, 288 stem fragments (48 per genotype) of *H. verticillata* were prepared, each consisting of a single-node stem fragment containing one peltate leaf, intact petiole, and two 1-cm internodes. This morphological standardization minimized size variation among individuals of the same genotype in the parental generation. Stem fragments were initially rooted in aquatic containers (52.5 cm long × 34.5 cm wide × 6.5 cm high) for 3 weeks, using a hydroponic solution of commercial liquid fertilizer (Scotts Miracle-Gro All Purpose; Scotts Miracle-Gro Co., Wuhan, China) diluted 1:400 (v/v) in tap water, refreshed weekly.

On July 27, 2022, rooted fragments were transplanted individually into pots (12 cm in diameter × 15.2 cm high), containing homogenized loess-sand substrate (1:1 v/v), yielding 288 potted plants. Plants were watered to saturation every 2 days during the 3-week establishment period.

On August 17, 2022, plants were transferred to nylon mesh cages (180 cm long × 60 cm wide × 60 cm high; 0.5 mm mesh). Each cage was divided into three compartments (60 cm long × 60 cm wide × 60 cm high) by transparent partitions (Fig. 1B). A total of 48 independent compartments were used, with six different genotypes planted in each compartment. Treatments (control or herbivory) were randomly assigned to cages, with all plants within the same cage receiving the same

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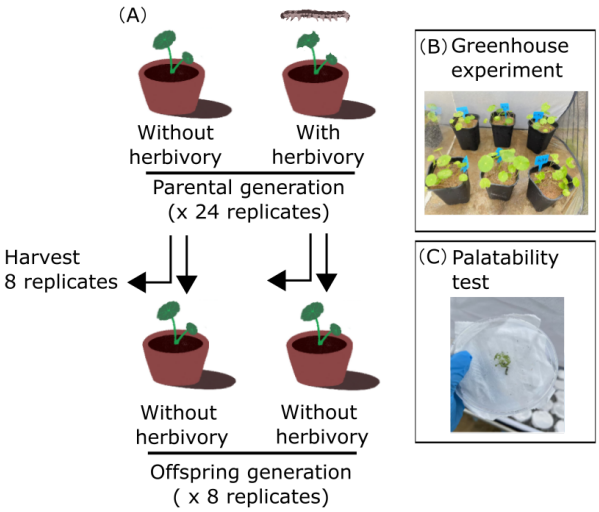

Fig. 1. Experimental design. There is a two-generation experimental design (A). In the parental generation, six genotypes of *Hydrocotyle verticillata* were grown in a greenhouse. For each genotype, half of the plants were exposed to herbivory by larvae of *Spodoptera litura*, while the other half were left unexposed to serve as controls. In the offspring generation, half of the clonal offspring are produced by the parental plants that had experienced herbivory, and the remaining half from parental plants that had not. All offspring were grown under no herbivory (B). One day before harvest, a mature leaf from each plant was collected for a palatability assay (C). To evaluate each leaf's susceptibility to herbivory, the harvested leaves were placed in Petri dishes and exposed to larvae of *Spodoptera litura* for 24 h. Panel B shows one of the three compartments within a cage, and Panel C shows the petri dish used in the palatability assay.

treatment. Within each compartment, the arrangement of plant genotypes was randomized to minimize spatial variation. This resulted in 24 replicates per treatment and genotype (6 genotypes × 2 treatments × 24 replicates).

In the parental generation, control plants remained herbivore-free, while herbivory treatment plants received four 2nd-instar *S. litura* larvae per pot. The choice of larvae density was based on preliminary trials to impose moderate but non-lethal defoliation, consistent with previous studies (*e.g.*, Dong *et al.* 2019; Fu *et al.* 2023). To maintain standardized herbivory pressure, larvae were introduced on August 17–23 and replaced with fresh second-instar larvae on August 24–30, 2022.

The parental generation experiment concluded on 31 August 2022. Eight randomly selected replicates per genotype × treatment combination were harvested. For each plant, we recorded the number of ramets, number of leaves, and stem length. Leaf area was measured using WinFOLIA software (Regent Instruments Inc., Quebec City, QC, Canada). Plant tissues (leaves, petioles, stems, and roots) were separately oven-dried at 70 °C for 48 h and weighed.

Offspring generation

On 1 September 2022, the remaining 16 replicates per genotype × treatment combination were used to evaluate the transgenerational effects. From each parental plant, a ramet from the second or third youngest stem node was excised at the internode using sterilized scissors, weighed, and transplanted individually. The clipping procedure was standardized to ensure consistent mechanical damage across all ramets while minimizing confounding effects from handling. Each ramet consisted of a single-node stem fragment with a peltate leaf, petiole, and two 1-cm internodes. A total of 192 ramets (6 genotypes × 2 parental treatments × 16 replicates) were established in homogenized loess-sand substrate (1:1 v/v). All offspring ramets remained herbivore-free for 11 weeks with watering every 2 days. Pots within each compartment were randomly repositioned weekly to minimize environmental heterogeneity.

The offspring experiment concluded on 19 November 2022. For each plant, we recorded survival status, ramet number, and stem length. One leaf from the second or third youngest node was collected for palatability testing (Fig. 1C). Remaining leaves were counted and measured for area. All tissues were oven-dried at 70°C for 48 h and weighed.

Palatability test

On November 20, 2022, a 24-h palatability test was conducted using 1st-instar *S. litura* larva that were pre-starved for 24 h in individual Petri dishes (10 cm diameter) at 21.26 °C ± 1.41 °C (Mean ± SE). Fresh leaves from offspring plants, trimmed to exclude petioles, were weighed and measured for area using "ImageJ" software (Schneider *et al.* 2012). After 24 h, uneaten material was re-measured to calculate herbivory-induced loss rates (Equation 1).

$$\text{Loss rate}(\%) = \frac{\text{pre} - \text{assay weight/area} - \text{post} - \text{assay weight/area}}{\text{pre} - \text{assay weight/area}} \times 100\% \quad (1)$$

Data analysis

Prior to analysis, we calculated mean leaf area (total leaf area/number of leaves), specific leaf area (SLA; total leaf area/total leaf mass), and root-to-shoot ratio (root mass/shoot mass).

Effects of genotype and herbivory in parental plants

We conducted linear mixed-effects models (LMMs) to evaluate the effects of genotypes, herbivory, and their interaction on growth (total mass, leaf mass, stem mass, petiole mass, root mass, number of ramets, number of leaves, stem length, and total leaf area), morphology (mean leaf area and SLA), and biomass allocation (root-to-shoot ratio) of parental plants. In the models, genotype and herbivory were treated as fixed factors, and cage was treated as a random factor to account for non-independence among plants within the same cage. LMMs were conducted using the 'lmer' function in the 'lme4' package (version 1.1.37; Bates *et al.* 2003). Type III ANOVAs were performed using the 'anova' function in the 'lmerTest' package (version 3.1.3; Kuznetsova *et al.* 2017). When main or interaction effects were significant, *post hoc* pairwise comparisons were conducted via the 'emmeans' and 'pairs' function in the 'emmeans' package (Lenth 2024).

Transgenerational effects on offspring survival and development

We used the binomial generalized linear model (GLM) to examine the effect of genotypes and PH on offspring survival. A complementary log–log (cloglog) link function was included

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in the model because the binomial response variable was asymmetric, with a substantially higher proportion of ones than zeros (Zuur *et al.* 2009). Model significance was evaluated via likelihood ratio tests (Morrell 1998). *Post hoc* comparisons were followed when main or interaction effects were significant.

For other traits, linear mixed-effects models (LMMs) with genotype and PH as fixed factors and cage as a random factor were conducted to test their effects on growth, morphology, and biomass allocation of clonal offspring. Analyses of offspring data followed the same workflow as described for parental analyses.

Transgenerational effects on offspring herbivory defence

We used quasi-binomial generalized linear models (GLMs) to examine the effect of genotype and PH on leaf loss ratio in clonal offspring. Herbivory defence was estimated using leaf loss ratio, where a lower ratio indicates the plant's ability to minimize leaf biomass loss under herbivore pressure. This approach was chosen to account for overdispersion in proportional response variables (Zuur *et al.* 2009). Model significance was evaluated via likelihood ratio tests, with *post hoc* comparisons using Holm correction.

All analyses were performed using R version 4.4.3 (R Core Team 2025).

RESULTS

Direct effects of genotype and herbivory on parental plants

Genotype significantly influenced the growth, morphology, and biomass allocation strategies of parental plants of *H. verticillata* (Tables 1 and 2). Genotype G4 had more vigorous growth and exhibited significantly greater total mass, leaf mass, petiole mass, and stem mass than most other genotypes (Fig. 2A–D). In contrast to its large above-ground biomass, G4 exhibited the lowest root-to-shoot ratio among all genotypes (Fig. 2F). G4 produced a low number of ramets and leaves but had the largest mean leaf area (Fig. 3A,D,E), so that G4 had a greater total leaf area than most other genotypes (Fig. 3C). Genotype G3 also exhibited significantly greater leaf mass, stem mass, and root mass than most other genotypes (Fig. 2B,D,E). Unlike G4, G3 displayed a different pattern, that is, G3 produced significantly more ramets, longer stolons, and more leaves than most other genotypes (Fig. 3A,B,D).

Herbivory exerted significant but selective effects on growth of parental plants (Tables 1 and 2). Specifically, herbivory significantly reduced both leaf mass and leaf number (Figs 2B and 3D), while also tending to reduce ramet production (Fig. 3A). Conversely, plants responded to herbivory with a significant increase in specific leaf area (SLA), while mean leaf area also showed a tendency to increase (Fig. 3E,F). However, no significant genotype × herbivory interactions were detected for any measured traits of parental plants.

Parental herbivory effects on offspring growth

Genotype significantly influenced the leaf number, morphology, and biomass allocation of clonal offspring, but did not broadly affect overall growth traits (Tables 3 and 4). Genotype G2 had a higher root-to-shoot ratio, and G4 produced the largest mean leaf area and had the highest SLA among all genotypes (Figs 4F and 5E,F). However, G4 also produced

Table 1. Results of linear mixed-effects model (LMM) analysis for effects of genotypes and insect herbivory, and their interaction on the growth of parental plants of *Hydrocotyle verticillata*. Type III ANOVAs were used to test the significance of fixed effects.

traits	effects	df	*F*	*P*-values
Total mass	Genotype (G)	5	**9.956**	**<0.001**
	Herbivory (H)	1	2.545	0.133
	G × H	5	1.440	0.221
Leaf mass	Genotype (G)	5	**7.154**	**<0.001**
	Herbivory (H)	1	**5.760**	**0.031**
	G × H	5	1.864	0.112
Stem mass	Genotype (G)	5	**9.284**	**<0.001**
	Herbivory (H)	1	0.710	0.414
	G × H	5	1.006	0.421
Petiole mass	Genotype (G)	5	**3.007**	**0.016**
	Herbivory (H)	1	3.071	0.101
	G × H	5	1.015	0.415
Root mass	Genotype (G)	5	**4.305**	**0.002**
	Herbivory (H)	1	1.428	0.252
	G × H	5	1.285	0.280
Number of ramets	Genotype (G)	5	**7.143**	**<0.001**
	Herbivory (H)	1	3.401	0.086
	G × H	5	2.207	0.063
Number of leaves	Genotype (G)	5	**14.576**	**<0.001**
	Herbivory (H)	1	**8.764**	**0.010**
	G × H	5	2.090	0.077
Stem length	Genotype (G)	5	**2.756**	**0.025**
	Herbivory (H)	1	1.555	0.233
	G × H	5	1.538	0.190
Total leaf area	Genotype (G)	5	**4.433**	**0.001**
	Herbivory (H)	1	0.058	0.814
	G × H	5	1.404	0.234

Significant effects ($P < 0.05$) are shown in bold. The number of replicates per treatment ranged from 7 to 8, as one G2 plant died unexpectedly in the herbivory treatment.

Table 2. Results of linear mixed-effects model (LMM) analysis for effects of genotypes and insect herbivory, and their interaction on the morphology and biomass allocation of parental plants of *Hydrocotyle verticillata*. Type III ANOVAs were used to test the significance of fixed effects.

traits	effects	df	*F*	*P*-values
Morphology				
Mean leaf area	Genotype (G)	5	**18.314**	**<0.001**
	Herbivory (H)	1	4.119	0.062
	G × H	5	1.146	0.344
Specific leaf area	Genotype (G)	5	1.302	0.274
	Herbivory (H)	1	**34.757**	**<0.001**
	G × H	5	1.380	0.243
Biomass allocation				
Root-to-shoot ratio	Genotype (G)	5	**4.159**	**0.002**
	Herbivory (H)	1	2.919	0.109
	G × H	5	0.897	0.488

Significant effects ($P < 0.05$) are shown in bold. The number of replicates per treatment ranged from 7 to 8, as one G2 plant died unexpectedly in the herbivory treatment.

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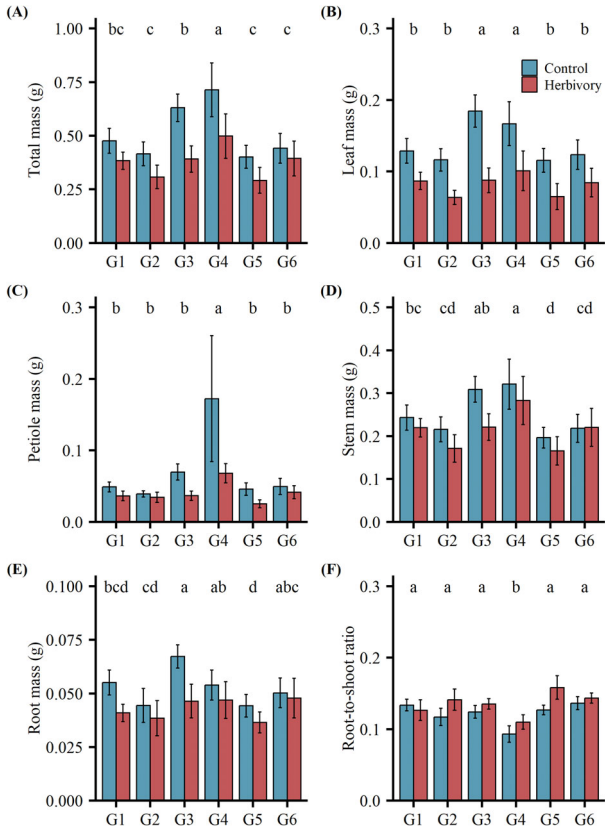

Fig. 2. Box plots showing effects of herbivory (control *versus* herbivory) and genotypes on biomass (A–E) and biomass allocation (root-to-shoot ratio; F) of *Hydrocotyle verticillata*. Different lowercase letters indicate significant differences among genotypes based on *post hoc* comparisons.

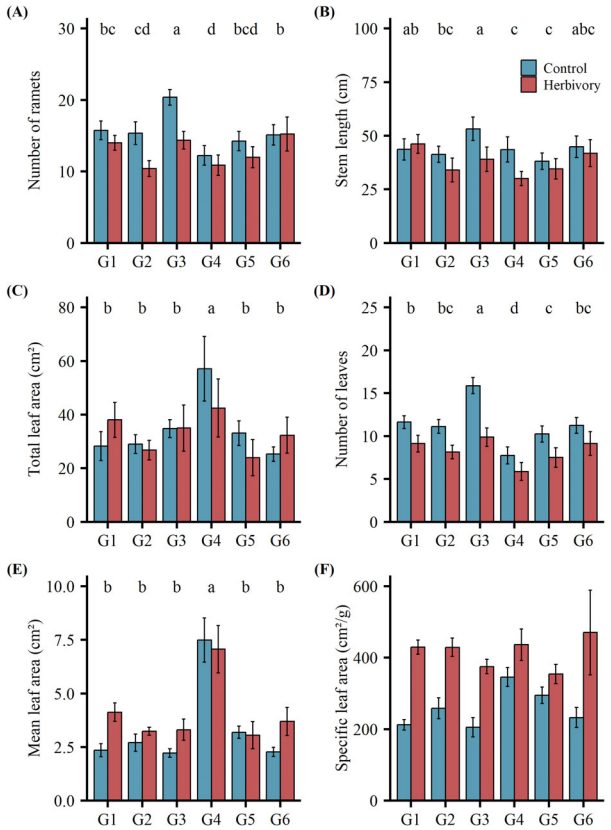

Fig. 3. Box plots showing effects of herbivory (control *versus* herbivory) and genotypes on growth (number of ramets, stem length, total leaf area, and number of leaves; A–D) and morphological traits (mean leaf area and specific leaf area; E, F) of parental plants of *Hydrocotyle verticillata*. Different lowercase letters indicate significant differences among genotypes based on *post hoc* comparisons.

significantly fewer leaves compared to all other genotypes (Fig. 5D). In contrast, G3 produced fewer leaves than G1 and G6 and also exhibited a smaller mean leaf area than genotypes G1 and G4 (Fig. 5D,E).

PH also tended to increase leaf mass and SLA of offspring plants (Tables 3 and 4; Figs 4B and 5F). Among all measures, the significant genotype × parental herbivory interactions were only detected for root mass and SLA, indicating that genotypic responses to parental cues were not uniform in some traits (Table 4). PH significantly reduced root mass in genotype G2 but had no significant effect on other genotypes (Fig. 4E). In contrast, PH significantly increased SLA in genotype G4 but had no significant effect on other genotypes (Fig. 5F).

It is also noteworthy that when accounting for variation in initial weight, the significant effects on leaf morphology (mean leaf area and SLA) remained unchanged, whereas genotypic effects on leaf number and root-to-shoot ratio, as well as herbivory effects on root mass, were reduced (Table S2). This suggests that the initial size of clonal offspring is involved in mediating parental effects.

Parental herbivory effects on offspring survival and defence

PH enhanced significantly the survival rate of clonal offspring ($P = 0.004$). Specifically, offspring produced by herbivory-treated parental plants exhibited a 21% higher survival rate compared to those from control parental plants (Fig. S2). Offspring genotype also significantly influenced survival, and genotype G4 had the lowest survival rate among all genotypes ($P < 0.001$). However, the effect of PH was independent of genotype, as indicated by the lack of a significant genotype × parental herbivory interaction ($P = 0.544$).

In the subsequent palatability test, the effect of PH did not extend to offspring palatability. The rate of leaf loss, measured as both leaf fresh mass and area, was significantly affected by genotype ($P < 0.001$ for mass and $P = 0.010$ for area), but not by PH or its interaction with genotype (all $P > 0.05$). Indeed, herbivores resulted in a substantial and statistically similar proportion of leaf tissue regardless of parental treatment, with average leaf mass reductions of 95.3% and 95.7%, and leaf area reductions of 81.2% and 83.2% for offspring from control and herbivory-treated parental plants, respectively (Fig. S3).

DISCUSSION

Direct effects of herbivory on parental plants

Above-ground herbivory by *S. litura* larvae seriously impaired the leaf development of *H. verticillata* (Fu *et al.* 2024). The reduced growth caused by *S. litura* is also consistent with

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Table 3. Results of linear mixed-effects model (LMM) analysis for effects of genotypes and parental herbivory, and their interaction on the growth of offspring of *Hydrocotyle verticillata*. Type III ANOVAs were used to test the significance of fixed effects.

traits	effects	df	*F*	*P*-values
Total mass	Genotype (G)	5	0.649	0.663
	Parental herbivory (PH)	1	0.376	0.557
	G × PH	5	1.404	0.227
Leaf mass	Genotype (G)	5	1.668	0.147
	Parental herbivory (PH)	1	4.399	0.057
	G × PH	5	1.105	0.361
Stem mass	Genotype (G)	5	0.737	0.597
	Parental herbivory (PH)	1	0.435	0.530
	G × PH	5	0.779	0.567
Petiole mass	Genotype (G)	5	1.674	0.145
	Parental herbivory (PH)	1	0.697	0.423
	G × PH	5	1.246	0.292
Root mass	Genotype (G)	5	1.865	0.105
	Parental herbivory (PH)	1	1.434	0.272
	G × PH	5	**2.552**	**0.031**
Number of ramets	Genotype (G)	5	0.920	0.471
	Parental herbivory (PH)	1	2.227	0.172
	G × PH	5	0.916	0.473
Number of leaves	Genotype (G)	5	**2.932**	**0.015**
	Parental herbivory (PH)	1	0.282	0.596
	G × PH	5	1.223	0.302
Stem length	Genotype (G)	5	1.264	0.284
	Parental herbivory (PH)	1	0.137	0.721
	G × PH	5	0.634	0.674
Total leaf area	Genotype (G)	5	2.177	0.061
	Parental herbivory (PH)	1	1.295	0.285
	G × PH	5	1.579	0.170

Significant effects ($P < 0.05$) are shown in bold. The number of replicates per treatment ranged from 4 to 16, as some plants died unexpectedly during the experiment.

Table 4. Results of linear mixed-effects model (LMM) analysis for effects of genotypes and parental herbivory, and their interaction on the morphology and biomass allocation of offspring of *Hydrocotyle verticillata*. Type III ANOVAs were used to test the significance of fixed effects.

traits	effects	df	*F*	*P*-values
Morphology				
Mean leaf area	Genotype (G)	5	**6.963**	**<0.001**
	Parental herbivory (PH)	1	1.354	0.280
	G × PH	5	1.303	0.267
Specific leaf area	Genotype (G)	5	**3.329**	**0.007**
	Parental herbivory (PH)	1	3.877	0.087
	G × PH	5	**4.248**	**0.001**
Biomass allocation				
Root-to-shoot ratio	Genotype (G)	5	**2.449**	**0.037**
	Parental herbivory (PH)	1	1.117	0.378
	G × PH	5	1.436	0.216

Significant effects ($P < 0.05$) are shown in bold. The number of replicates per treatment ranged from 4 to 16, as some plants died unexpectedly during the experiment.

previous evidence from Fu *et al.* (2024) and supports the first hypothesis. In response to such herbivory, *H. verticillata* produced larger and thinner leaves. This morphological plasticity

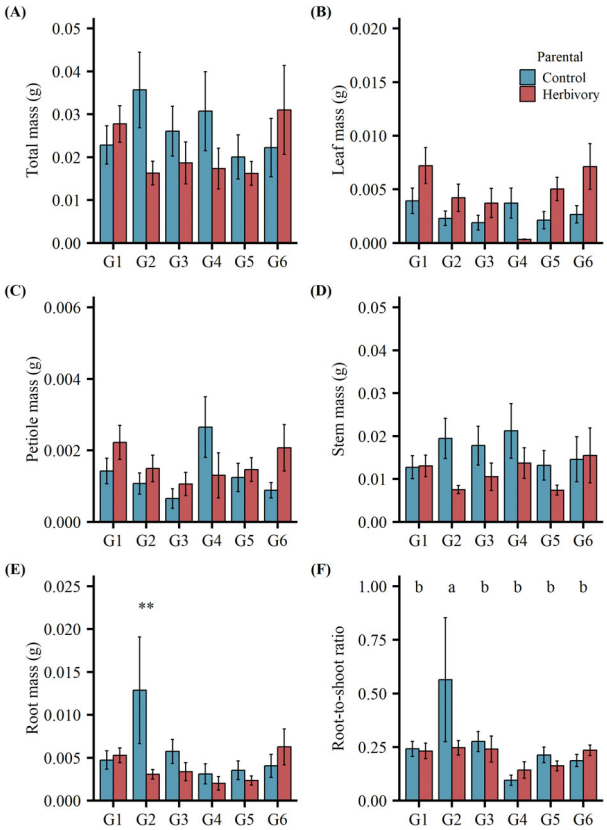

Fig. 4. Box plots showing effects of parental herbivory (control *versus* herbivory) and genotypes on biomass (A–E) and biomass allocation (root-to-shoot ratio; F) of offspring plants of *Hydrocotyle verticillata*. Different lowercase letters indicate significant differences among genotypes based on *post hoc* comparisons.

likely represents an adaptive strategy, because larger leaves could mitigate herbivory damage across a greater surface area, by minimizing proportional losses to photosynthetic tissues (Cunha *et al.* 2023). Thinner leaves may optimize light capture and gas exchange efficiency, which could potentially offset carbohydrate deficits caused by herbivory damage (Hanba *et al.* 2023). Notably, while this morphological plasticity may offer compensatory benefits, these larger but thinner leaves may also increase plant apparency and vulnerability to subsequent herbivore attacks (Garibaldi *et al.* 2011). Therefore, this trade-off between compensatory growth and increased herbivore susceptibility represents a fundamental dilemma for *H. verticillata* under herbivore pressure, where immediate recovery benefits must be balanced against future vulnerability.

The substantial genotypic variation observed across all measured traits underscores how genetic background fundamentally governs plant development and phenotypic expression in *H. verticillata*. In particular, genotype G4's consistent expression of higher total biomass, expanded leaf area, and reduced root-to-shoot ratio contrasts markedly with G3's strategy of enhanced ramet production and increased leaf number. These divergent trait syndromes reveal fundamentally different resource allocation strategies among genotypes, demonstrating that genetic background profoundly influences phenotypic expression patterns (Solé-Medina *et al.* 2022).

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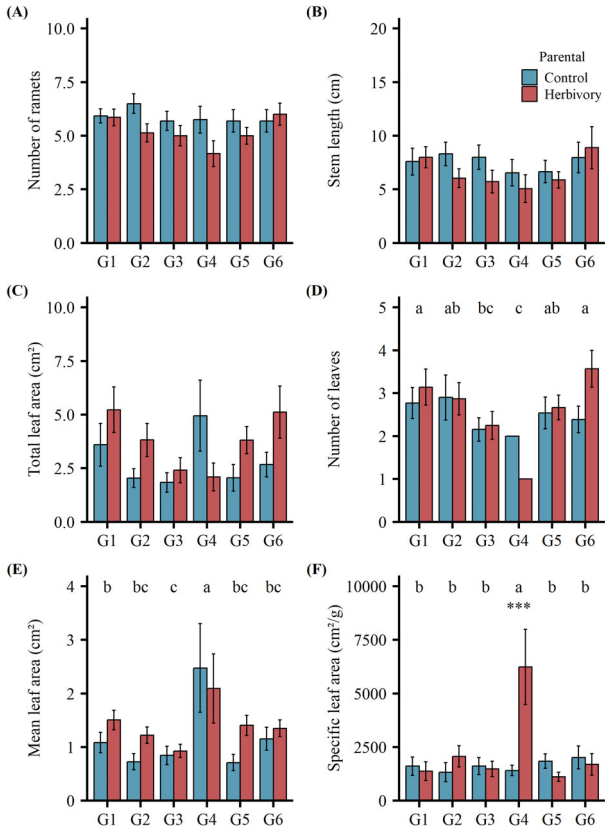

Fig. 5. Box plots showing effects of parental herbivory (control *versus* herbivory) and genotypes on growth (number of ramets, stem length, total leaf area, and number of leaves; A–D) and morphological traits (mean leaf area and specific leaf area; E, F) of offspring plants of *Hydrocotyle verticillata*. Different lowercase letters indicate significant differences among genotypes based on *post hoc* comparisons.

On the other hand, the absence of significant genotype-by-herbivory interactions indicates that herbivory effects on leaf traits of *H. verticillata* are relatively uniform across genetic backgrounds. However, the maintenance of distinct genotypic signatures regardless of herbivory treatment also reveals that genetic constraints override environmental plasticity in determining morphological outcomes (Nicotra *et al.* 2010). This genetic determinism may confer selective advantages, as some genotypes that maintain superior performance across environmental gradients are better positioned to exploit heterogeneous habitats – a key attribute facilitating invasive success (Richards *et al.* 2006; Palacio-López & Gianoli 2011).

Parental herbivory effects on offspring survival and development

PH induced selective but adaptive transgenerational effects in clonal offspring of *H. verticillata*. Most notably, offspring survival increased by 21% when derived from herbivory-treated parental plants, demonstrating adaptive transgenerational priming despite the limited scope of morphological modifications. Among growth and morphological traits, only leaf mass and SLA exhibited an increasing trend in clonal offspring from herbivory-treated parental plants, indicating targeted rather than systemic resource reallocation. This contrasts with findings from another clonal invasive plant, *Alternanthera philoxeroides* (Mart.) Griseb., which showed transgenerational effects in most growth traits (Fu *et al.* 2023). This selective increase in photosynthetic tissue in the absence of total biomass changes suggests targeted resource reallocation rather than generalized growth enhancement (Burow & Halkier 2017). Such targeted modifications may exemplify anticipatory transgenerational plasticity, where parental stress exposure calibrates offspring phenotypes for improved performance under similar environmental challenges (Wadgymar *et al.* 2018; Xiao *et al.* 2023). These findings partially support our second hypothesis regarding transgenerational effects through clonal reproduction, aligning with observations in other clonal systems where herbivory induces similar offspring modifications (Dong *et al.* 2018b, 2019).

The genotype-specific patterns in root mass and SLA responses further illuminate the complexity of transgenerational plasticity. While most genotypes maintained relatively stable traits regardless of parental treatment, genotype G2 showed a marked decrease in root mass and G4 showed an increase in SLA following PH, suggesting lineage-specific stress memory mechanisms. This contrasting response pattern indicates that transgenerational effects operate through both conserved pathways (affecting survival and leaf mass across all genotypes) and genotype-dependent mechanisms (modulating root mass and SLA in specific lineages). Moreover, the enhanced survival despite limited morphological changes suggests that transgenerational priming may operate largely through some cryptic physiological or biochemical modifications not captured by standard morphological assessments (Sobral *et al.* 2021; Louis *et al.* 2023).

Parental herbivory effects on offspring defence

The complete absence of enhanced herbivore defence in clonal offspring contradicts established expectations of defensive transgenerational priming (*e.g.*, Mesa *et al.* 2019) and refutes part of our second hypothesis. Despite improved survival rates, offspring from herbivory-exposed parents suffered catastrophic leaf losses (95.3%) statistically indistinguishable from control offspring (95.7%), revealing a fundamental decoupling of survival enhancement from defence induction. This paradoxical outcome – improved survival without defensive capability – suggests that transgenerational effects in *H. verticillata* operate primarily through growth optimization rather than defence mechanisms.

This asymmetric transgenerational response aligns with the Enemy Release Hypothesis while revealing its nuances. Our results demonstrate that invasive populations not only redirect constitutive resources from defence to growth (Salgado-Luarte *et al.* 2023; Yi *et al.* 2024) but also fail to maintain inducible transgenerational defence mechanisms. The enhanced offspring survival and leaf mass allocation occur without any corresponding defensive memory, creating populations optimized for rapid growth but remaining entirely vulnerable to herbivory. Indeed, genotype G4 exemplifies this trade-off: despite showing the strongest growth responses and highest SLA, it exhibited the lowest survival rates, indicating that extreme growth orientation may compromise stress resilience. For invasive clonal plants, this evolutionary strategy may facilitate initial colonization success but could prove maladaptive if herbivore communities reassemble in invaded ranges (Agrawal *et al.* 1999; van Kleunen *et al.* 2010; Züst & Agrawal 2017).

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Genotypic differences in parental effects

The emergence of genotype-specific transgenerational responses reveals complex interactions between genetic architecture and environmental memory in *H. verticillata*. Our analysis identified two distinct transgenerational strategies. Most genotypes (*e.g.*, G1, G2, G3, G5, and G6) exhibited a fixed transgenerational strategy characterized by phenotypic canalization – maintaining stable trait expression across parental environments with minimal plasticity. Conversely, genotype G4 demonstrated a conditional strategy, displaying environment-dependent phenotypic modifications, particularly herbivory-induced increases in SLA. These contrasting strategies indicate that transgenerational plasticity is not uniformly distributed across genotypes but rather reflects underlying genetic variation in stress-response pathways (Agrawal *et al.* 1999; Herman & Sultan 2011; Holeski *et al.* 2012).

Genotype G4's response pattern particularly exemplifies the complexity of transgenerational inheritance. Its maintenance of elevated SLA and expanded leaf area following PH parallels genotype-dependent stress-priming documented in *Arabidopsis thaliana*, where epigenetic and metabolic inheritance mechanisms generate lineage-specific responses (Latzel *et al.* 2023). This phenotypic modification might represent either an evolved bet-hedging strategy providing resilience against environmental unpredictability or the expression of pre-existing adaptive networks optimized for resource acquisition. However, G4's paradoxically low offspring survival reveals critical growth-survival trade-offs, demonstrating that enhanced growth potential does not necessarily translate to improved survival outcomes.

These patterns strongly support our third hypothesis regarding genotypic variation in parental effects. The distinction between genotype-independent responses (such as the generally increased leaf mass) and genotype-dependent responses (like G4's unique SLA increase) illuminates the dual nature of transgenerational plasticity. While shared physiological mechanisms may provide broad environmental buffering, genotype-specific responses enable fine-tuned adaptation to local selective pressures, potentially driving evolutionary divergence within invasive populations (Eller *et al.* 2017; Garthwaite *et al.* 2024).

CONCLUSIONS

Our findings reveal the nuanced transgenerational dynamics underlying invasion success in *H. verticillata*, demonstrating how clonal plants balance growth enhancement against defensive abandonment. While PH enhanced offspring survival by 21% and tended to increase the investment of leaf mass, it failed entirely to induce herbivory defence, with offspring suffering catastrophic defoliation (>95%) regardless of parental treatment. This asymmetric response – enhanced growth provisioning without defensive memory – represents a fundamental trade-off in invasive populations freed from co-evolved herbivore pressure. The decoupling of survival enhancement from defence induction challenges conventional expectations of transgenerational priming and highlights an evolutionary gamble: maximizing colonization potential through growth-oriented resource provisioning while sacrificing induced defence capacity.

In addition, genotypic variation profoundly modulated these transgenerational effects, revealing both conserved and lineage-specific response pathways. While enhanced survival and the potential of increased leaf mass occurred across most genotypes, suggesting shared physiological mechanisms, genotype G4 exemplified an alternative strategy with dramatically elevated SLA responses to PH yet paradoxically lowest offspring survival. This genotype-dependent heterogeneity indicates that invasion success may depend not on uniform population responses but on maintaining diverse transgenerational strategies within populations. Understanding these genotype-specific mechanisms and their ecological consequences is essential for predicting invasion trajectories and developing targeted management strategies that exploit the defensive gaps revealed by our study, particularly as herbivore communities reassemble in invaded ranges.

AUTHOR CONTRIBUTIONS

B-CD and F-HY conceived the idea and supervised the study. Q-YF conducted the experiments and collected the data. Y-FZ, H-CC, Q-YF, and B-CD performed the analyses. Y-FZ and Q-YF wrote the initial draft of the manuscript. All authors contributed to editing the manuscript.

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CONFLICT OF INTEREST

The authors declare no competing interests.

DATA AVAILABILITY STATEMENT

Data and R codes are publicly available in the GitHub repository (https://github.com/ZhaoYingfei/parental_effects_and_genotypes).

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Data S1.

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